

| VECTORS        |                                                                                                                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Term           | Definition                                                                                                                                                                                                                                                                                                                       |
| Norm/Length    | $\ v\  = \sqrt{\sum_{i=1}^n v_i^2}$ $\ v\  = \sqrt{v^T v}$ $\nabla \ v\  = \frac{1}{\ v\ } v^T$                                                                                                                                                                                                                                  |
| Distance       | $d(u, v) = \ u - v\ $ <div><b>TODO:</b><math display="block">\nabla d(u, v) = \frac{1}{d(u, v)}((u - v)(u - v))^{??}</math><math display="block">\nabla_u d(u, v) = \frac{u - v}{\ u - v\ } = \frac{u - v}{d(u, v)}</math><math display="block">\nabla_v d(u, v) = -\frac{u - v}{\ u - v\ } = \frac{v - u}{d(u, v)}</math></div> |
| Scalar product | $u \bullet v = u^T v = \sum_k u_k v_k$                                                                                                                                                                                                                                                                                           |

SCALAR PRODUCT

MATRICES

Determinant:

$$\det(a) = a$$
$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$-a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5)$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$
$$= \begin{pmatrix} (2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8) & (2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9) \\ (4 \cdot 6 + -1 \cdot 1 + 7 \cdot 8) & (4 \cdot 4 + -1 \cdot 0 + 7 \cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

INDICES

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, A \cdot B \in \mathbb{R}^{n \times r}$$
$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$$
$$((A \cdot B)^T)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ik}$$
$$a^T \cdot B \cdot a = \sum_{ij} (a^T)_i B_{ij} a_j = \sum_{ij} a_i B_{ij} a_j$$
$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$$
$$(E_{rst})_{ijk} = \delta_{ri} \delta_{sj} \delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$
$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M \cdot x$$

$$L_A(\vec{0}) = \vec{0}$$

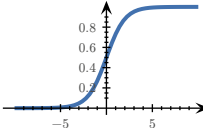
**DERIVATIVES**

|                                                                 |                                                             |
|-----------------------------------------------------------------|-------------------------------------------------------------|
| $1 \rightarrow 0$                                               | $x \rightarrow 1$                                           |
| $x^2 \rightarrow 2x$                                            | $x^n \rightarrow nx^{n-1}$                                  |
| $\frac{1}{x} \rightarrow -\frac{1}{x^2}$                        | $\sqrt{x} \rightarrow \frac{1}{2\sqrt{x}}$                  |
| $e^x \rightarrow e^x$                                           | $e^{-x} \rightarrow -e^{-x}$                                |
| $\ln(x) \rightarrow \frac{1}{x} \text{ for } x > 0$             | $\ln(y \cdot x) \rightarrow \frac{1}{x} \text{ for } x > 0$ |
| $a^x \rightarrow \ln(a) \cdot a^x \text{ for } a > 0, a \neq 1$ | $\log_a(x) \rightarrow \frac{1}{\ln(b) \cdot x}$            |
| $\sin(x) \rightarrow \cos(x)$                                   | $\sin(2x) \rightarrow 2 \cos(2x)$                           |
| $\cos(x) \rightarrow -\sin(x)$                                  | $\cos(ax) \rightarrow -a \sin(ax)$                          |
| $\tan(x) \rightarrow 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$       | $\operatorname{arccot}(x) \rightarrow -\frac{1}{1+x^2}$     |
| $\cot(x) \rightarrow -\frac{1}{\sin^2(x)}$                      | $\arcsin(x) \rightarrow \frac{1}{\sqrt{1-x^2}}$             |
| $\operatorname{arccos}(x) \rightarrow -\frac{1}{\sqrt{1-x^2}}$  | $\arctan(x) \rightarrow \frac{1}{1+x^2}$                    |

| Term           | Definition                                                                                                                     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| Addition       | $(f(x) + g(x))' = f'(x) + g'(x)$                                                                                               |
| Multiplication | $(\alpha \cdot f(x))' = \alpha \cdot f'(x)$                                                                                    |
| Product rule   | $(f \cdot g)' = f' \cdot g + f \cdot g'$ $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$ |
| Chain rule     | $(f(g(x)))' = f'(g(x)) \cdot g'(x)$                                                                                            |
| Quotient rule  | $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$                                                                            |

FUNCTIONS OF SEVERAL VARIABLES

| Term                           | Definition                               |
|--------------------------------|------------------------------------------|
| Real valued function           | $R \subset \mathbb{R}$                   |
| Vector valued function         | $R \subset \mathbb{R}^m$                 |
| A function of $n$ variables    | $D \subset \mathbb{R}^n$                 |
| Softmax function               | <b>TODO:</b>                             |
| RSS                            | $RSS = \sum_i (y_i - f(x_i))^2$          |
| $n$ -dimensional curve         | $c: \mathbb{R} \rightarrow \mathbb{R}^n$ |
| $n$ -dimensional hyper-surface | $s: \mathbb{R}^n \rightarrow \mathbb{R}$ |
| Reparametrization              | $c(t) \rightarrow d(s) = c(h(s))$        |

**Sigmoid function:**
$$\sigma(x) : \mathbb{R} \rightarrow [0; 1] = \frac{1}{1 + e^{-x}}$$


DERIVATIVES

| Function                                                                                                                       | Derivative                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $f: \mathbb{R} \rightarrow \mathbb{R}^n$ $f: \begin{cases} \mathbb{R} \mapsto y \end{cases}$                                   | $f': \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ \mathbb{R} \mapsto y \end{cases} = \begin{pmatrix} \frac{d}{dx} f_1(x) \\ \frac{d}{dx} f_2(x) \\ \vdots \\ \frac{d}{dx} f_n(x) \end{pmatrix}$                                                                                                                                                                                                                                                               |
| $f: \mathbb{R}^n \rightarrow \mathbb{R}$ $f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R} \\ x \mapsto y \end{cases}$     | $f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{1 \times n} \\ x \mapsto y \end{cases} = \nabla f$ $= \left( \frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right)$                                                                                                                                                                                                                       |
| $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ $f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^m \\ x \mapsto y \end{cases}$ | $f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n} \\ x \mapsto y \end{cases} = J_f = \frac{\partial f}{\partial x} \Big _{x=x_0}$ $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix}$ |

PARTIAL VS TOTAL DERIVATIVE

**TODO:**  
example FS2024

$$f(x, y) = x^2 + 2y$$
$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{d f}{d x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{d y}{d x} = 2x + 2 \cdot \frac{d y}{d x} \mid (y \text{ dependent on } x)$$

GRADIENTS

Common gradients:

$$f(x) = \|x\|^2 \Rightarrow \nabla f = 2x$$
$$f(x) = a^T x + c \Rightarrow \nabla f = a$$
$$f(x) = x^T A x \Rightarrow \nabla f = (A + A^T) x$$
$$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow \nabla f = \frac{A + A^T}{2} x + b$$
$$= A x + b \text{ if } A^T = A$$
$$f(x) = g(A x + b) \Rightarrow \nabla f = A^T \nabla g(A x + b)$$
$$g: \mathbb{R}^m \rightarrow \mathbb{R}$$
$$f(x) = u(x) \cdot v(x) \Rightarrow \nabla f = (J_u)^T v + (J_v)^T u$$
$$f(x) = \|u(x)\| = \sqrt{u^T u} \Rightarrow \nabla f = (J_u)^T \frac{u}{\|u\|} \text{ if } \|u\| \neq 0$$
$$f(x) = \log p(x) \Rightarrow \nabla f = \frac{1}{p(x)} \nabla p(x)$$

JACOBIAN MATRIX

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2, J_f(x, y) = \begin{pmatrix} \frac{\partial x_1}{\partial x_2} & \frac{\partial y_1}{\partial x_2} \end{pmatrix}$$

Linearisation:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, x_0 \in \mathbb{R}^n$$
$$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$$

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}^k, h: \mathbb{R}^n \rightarrow \mathbb{R}^k = g \circ f$$
$$J_h = J_g(f(x)) \cdot J_f(x)$$

Common jacobians:

$$f(x) = a + M \cdot x \Rightarrow J_f = M$$
$$f(x) = x \Rightarrow J_f = \mathbb{1}$$
$$f(x) = \|x\|^2 \Rightarrow J_f = (2x)^T$$

**TODO:**  
 $J_x$  for softmax  $s_i = \frac{e^{x_i}}{\sum_k e^{x_k}}: J_{ij} = s_i(\delta_{ij} - s_j)$

HYPER-SURFACES

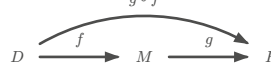
Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}, h: \mathbb{R}^n \rightarrow \mathbb{R} = g \circ f$$
$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f) |_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

Differentiation properties:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, g: \mathbb{R}^n \rightarrow \mathbb{R}$$
$$\nabla(f + g) = \nabla f + \nabla g$$
$$\nabla(f - g) = \nabla f - \nabla g$$
$$\nabla(f \cdot g) = g \nabla f + f \nabla g$$
$$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$$f: D \rightarrow M$$
$$g: N \rightarrow R$$
$$h: D \rightarrow R$$
$$= g \circ f$$


GEOMETRY OF HYPER-SURFACES

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

Tangent space:  $\alpha, \beta \in \mathbb{R}, T =$

$$\left\{ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$
$$\left\{ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Graph of a linearisation:

$$l(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) = T_{p_0}$$

SECOND DERIVATIVE

$$\frac{\partial}{\partial x} \left( \frac{\partial}{\partial x} (f(x, y)) \right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx} f(x, y)$$

HESSIAN MATRIX

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$
$$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$
$$(H(x))_{ij} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$$
$$\partial_i \partial_j f = \partial_j \partial_i f$$

Common Hessians:

$$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$$
$$f(x) = g(Ax + b) \Rightarrow H_f = A^T H_g(Ax + b) A$$
$$g: \mathbb{R}^m \rightarrow \mathbb{R}$$

EXTREMAL POINTS

Let  $z = f(c(t))$ ,  $c$  a curve passing the stationary point  $c_0 = (x_0, y_0)$  at  $t = t_0$  and

$$\frac{d^2 z}{dt^2} = \underbrace{(\dot{c}_1(t_0), \dot{c}_2(t_0))}_{v^T} \cdot \underbrace{\begin{pmatrix} f_{xx}(c_0) & f_{xy}(c_0) \\ f_{yx}(c_0) & f_{yy}(c_0) \end{pmatrix}}_H \cdot \underbrace{\begin{pmatrix} \dot{c}_1(t_0) \\ \dot{c}_2(t_0) \end{pmatrix}}_v$$

- 1) If  $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\} (v^T H v > 0)$ , then  $\frac{d^2 z}{dt^2} > 0$  defines a **local minimum**
- 2) If  $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\} (v^T H v < 0)$ , then  $\frac{d^2 z}{dt^2} < 0$  defines a **local maximum**
- 3) If  $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\} (v^T H v < 0 \vee v^T H v > 0)$ , the point  $c_0 = (x_0, y_0)$  defines a **saddle point**

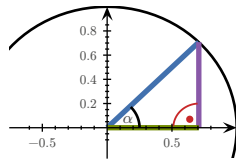
**TODO:**
$$(J_u)_{ij} = \frac{\partial u_i}{\partial x_j}, (H_g)_{ij} = \frac{\partial^2 g}{\partial x_i \partial x_j}$$
$$\partial_j (u_i u_i) = (\partial_j u_i) v_i + u_i (\partial_j v_i)$$

MISC

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

TRIGONOMETRY

| $x$       | 0 | $30^\circ = \frac{\pi}{6}$ | $45^\circ = \frac{\pi}{4}$ | $60^\circ = \frac{\pi}{3}$ | $90^\circ = \frac{\pi}{2}$ |
|-----------|---|----------------------------|----------------------------|----------------------------|----------------------------|
| $\sin(x)$ | 0 | $\frac{1}{2}$              | $\frac{\sqrt{2}}{2}$       | $\frac{\sqrt{3}}{2}$       | 1                          |
| $\cos(x)$ | 1 | $\frac{\sqrt{3}}{2}$       | $\frac{\sqrt{2}}{2}$       | $\frac{1}{2}$              | 0                          |
| $\tan(x)$ | 0 | $\frac{1}{\sqrt{3}}$       | 1                          | $\sqrt{3}$                 |                            |

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$
$$\sin(2x) = 2 \sin(x) \cos(x)$$
$$\sin^2(x) + \cos^2(x) = 1$$
$$\mathbf{1, \sin(\alpha), \cos(\alpha)}$$


GRADIENT DESCENT / NEWTON'S METHOD

$$GD: x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$$
$$N: x_{i+1} = x_i - \gamma \cdot (H_{f(x_i)})^{-1} \cdot \nabla f(x_i)$$
$$\gamma = \text{step size}$$
$$\text{termination criterion } \varepsilon > |x_{i+1} - x_i|$$

EXAMPLES

**TODO:**